

Aurélien Cécille

✉ cecille.aurelien@gmail.com

☎ +33 6 09 22 03 96

📍 Lyon, France

🌐 Aurel-C



Education

Ph.D. Candidate in Computer Science

LIRIS & Visual Behavior *May 2023 - June 2026*
Structure Aware Self-Supervised Monocular Depth Estimation

Master's Degree in Computer Science

University of Technology of Troyes *2017 - 2022*

Work Experience

Research Engineer

Visual Behavior *Oct. 2022 - Oct. 2025*

- Developed computer vision modules for intelligent vehicles and mobile robotics
- Estimated lead-vehicle distance and detected traffic lights for energy-aware driving applications
- Built local occupancy maps to support safe navigation around obstacles and people
- Contributed to a production ADAS system, with work on free-space and vulnerable road user detection

AI Engineer Intern

Orange *Feb. 2022 - Jul. 2022*

- Detected and segmented fiber optic cables to evaluate the quality of network closets
- Evaluated few-shot learning methods based on meta-learning
- Leveraged latent representations from generative models (GANs) to improve predictions in low-annotation settings

AI Engineer Intern

Visual Behavior *Sept. 2020 - Feb. 2021*

- Self-supervised learning for depth estimation, optical flow, and egomotion estimation
- Environment reconstruction from bird's-eye view

Projects

Conditional Logo Generation

- Logo generation using StyleGAN2
- Conditioning with latent-space clustering from a VAE
- Flask interface with color, shape, and category controls

Reinforcement Learning Agent

- Custom grid-world environment in Unity ML-Agents linked with OpenAI Gym
- Implementation in JAX for parallelization and optimized hyperparameter search

Research Interests

Computer Vision Deep Learning
Self-Supervised Learning Robotics

Publications

Towards Sharper Object Boundaries in Self-Supervised Depth Estimation

Aurélien Cécille, Stefan Duffner, Franck Davoine, Rémi Agier & Thibault Neveu
BMVC Oral, November 27, 2025, Sheffield, United Kingdom

GroCo: Ground Constraint for Metric Self-Supervised Monocular Depth

Aurélien Cécille, Stefan Duffner, Franck Davoine, Thibault Neveu & Rémi Agier
ECCV 2024, October 2024, Milan, Italy

Summer Schools & Workshops

ICVSS 2025

Computer Vision for Spatial Intelligence
Summer school attendance & poster

PAISS 2025

Summer school attendance

Naver Labs Workshop 2025

AI for Robotics
Attendance and poster presentation

Technical Skills

Core tools
PyTorch JAX Python OpenCV Docker
Linux Git OpenGL WebGPU D3.js

Languages

English : Fluent (C2 level)
French : Native language